University of Puerto Rico at Mayagüez

Department Of Chemical Engineering

Course: INQU 4001 Heat Transfer

Numerical Methods in Heat Conduction

Finite Difference Analysis and Thermal Optimization of a Composite Wall

Heat Transfer – Chapter 5, Problem 5-33

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Introduction

Heat transfer analysis plays an important role in engineering applications involving temperature control, material selection, and thermal system performance. Understanding how heat moves through different materials allows engineers to evaluate temperature distributions and improve thermal designs. This project focuses on Problem 5-33 from Chapter 5, Numerical Methods in Heat Conduction, which involves steady-state, one-dimensional heat transfer through a composite wall consisting of stainless steel and nonmetallic material. The original problem is solved using the finite difference method to determine the temperature distribution across nine nodes and evaluate the thermal performance of the composite wall. An adapted version of the problem is then developed by modifying the thermal conductivity of the nonmetallic layer and the convection heat transfer coefficient at the upper surface. These modifications are intended to reduce thermal resistance, increase the steady-state heat flux, and lower the operating temperatures of the materials. Artificial intelligence tools are used to assist with the numerical analysis, while manual calculations are performed to verify the results, identify potential errors, and ensure consistency with fundamental heat transfer principles.

Project Objectives

The main objective of this project is to evaluate and improve the thermal performance of a composite wall using the finite difference method and AI-assisted engineering analysis.

The specific objectives are:

- Solve the original Problem 5-33 using steady-state, one-dimensional heat conduction principles.
- Determine the temperature distribution across all nine nodes.
- Apply the appropriate convection boundary conditions and interface energy balance.
- Develop an adapted thermal configuration by modifying material conductivity and convection conditions.
- Compare the heat flux and temperature distributions of the original and adapted configurations.
- Independently verify AI-generated calculations and identify numerical or engineering inconsistencies.
- Document the AI prompts, outputs, revisions, and manual verification procedures using GitHub.

Original Problem Description

Problem 5-33 from Chapter 5, Numerical Methods in Heat Conduction, analyzes steady-state heat transfer through a composite wall consisting of two different materials: stainless steel ASTM A240 904L and a nonmetallic layer. The stainless-steel plate has a thickness of 30 mm and a thermal conductivity of 13 W/m·K, while the nonmetallic plate has a thickness of 10 mm and a thermal conductivity of 3 W/m·K. The bottom surface is exposed to steam at 260 °C with a convection heat transfer coefficient of 300 W/m²·K. The top surface is exposed to air at 20 °C with a convection heat transfer coefficient of 50 W/m²·K. The objective of the original problem is to determine the temperature distribution throughout the composite wall using the finite difference method with a uniform nodal spacing of 5 mm. The results are also evaluated against the temperature limits specified for the stainless steel and the ASTM B211 6061 bolts.

Given Data:

Table 1. Data from Original Problem

Parameter	Symbol	Value
Stainless steel thermal conductivity	k_1	13 W/m·K
Stainless steel thickness	L_1	0.030 m
Nonmetallic material conductivity	k_2	3 W/m·K
Nonmetallic layer thickness	L_2	0.010 m
Steam temperature	$T_{\infty,1}$	260 °C
Bottom convection coefficient	h_1	300 W/m²·K
Air temperature	$T_{\infty,2}$	20 °C
Top convection coefficient	h_2	50 W/m²·K
Nodal spacing	Δx	0.005 m
Stainless steel temperature limit	$T_{max,steel}$	260 °C
Bolt temperature limit	$T_{max,bolt}$	204 °C

Assumptions

The following assumptions are considered in the analysis:

1. Heat transfer occurs under steady-state conditions.
2. Heat conduction is one-dimensional.
3. Thermal conductivity remains constant within each material.
4. No internal heat generation occurs.

5. Thermal contact resistance between the materials is negligible.

6. Radiation heat transfer is neglected.

7. Heat conduction through the bolts is not explicitly modeled.

These assumptions simplify the physical system and allow the temperature distribution to be determined using a one-dimensional finite difference formulation.

Finite Difference Methodology

The finite difference method (FDM) was applied to determine the steady-state temperature distribution across the composite wall. This numerical technique approximates the governing heat conduction equation by dividing the physical domain into discrete nodes and applying energy balances at each location.

Table 2. Nodal positions and material identification for the composite wall.

Node	Position, x (mm)	Material/location
0	0	Bottom surface of stainless steel
1	5	Stainless steel
2	10	Stainless steel
3	15	Stainless steel
4	20	Stainless steel
5	25	Stainless steel
6	30	Interface between both materials
7	35	Nonmetallic material
8	40	Top surface of nonmetallic material

For this problem, the total wall thickness is 40 mm, consisting of 30 mm of stainless steel and 10 mm of nonmetallic material. A uniform nodal spacing of 5 mm was used, resulting in nine nodes numbered from 0 to 8. The coordinate system was defined from the bottom stainless-steel surface, exposed to hot steam, toward the upper nonmetallic surface, exposed to air. Node 6 represents the interface between both materials.

Governing Equation

For one-dimensional, steady-state conduction without internal heat generation and with constant thermal conductivity, the governing equation is:

$$\frac{dT^2}{dx^2} = 0$$

Using the central finite difference approximation, the governing equation becomes:

$$\frac{T_{i-1} - 2T_i + T_{i+1}}{(\Delta x)^2} = 0$$

Therefore, the finite difference equation for an interior node within a homogeneous material is:

$$T_{i-1} - 2T_i + T_{i+1} = 0$$

Boundary Conditions

At the bottom surface (node 0), heat is transferred from the steam to the stainless steel by convection. The energy balance is:

$$h_1(T_{\infty,1} - T_0) = \frac{k_1}{\Delta x}(T_0 - T_1)$$

Substituting the original problem parameters:

$$300(260 - T_0) = \frac{13}{0.005}(T_0 - T_1)$$

Rearranging:

$$2{,}900T_0 - 2{,}600\,T_1 = 78{,}000$$

At the upper surface (node 8), heat is transferred from the nonmetallic material to the surrounding air by convection:

$$\frac{k_2}{\Delta x}(T_7 - T_8) = h_2(T_8 - T_{\infty,2})$$

Substituting:

$$\frac{3}{0.005}(T_7 - T_8) = 50(T_8 - T_{\infty,2})$$

Rearranging:

$$-600T_7 + 650T_8 = 1{,}000$$

Material Interface

Node 6 represents the interface between stainless steel and nonmetallic material. Because the materials have different thermal conductivities, an energy balance is applied to maintain heat flux continuity:

$$\frac{k_2}{\Delta x}(T_5 - T_6) = \frac{k_2}{\Delta x}(T_6 - T_7)$$

Substituting the original conductivities:

$$2600(T_5 - T_6) = 600(T_6 - T_7)$$

Therefore:

$$2{,}600T_5 - 3{,}200T_6 + 600T_7 = 0$$

System of Linear Equations

The finite difference equations were assembled into a system of nine linear equations corresponding to the nine unknown nodal temperatures. The system can be expressed in matrix form as:

$$AT = b$$

where $\boldsymbol{A}$ is the coefficient matrix, $\boldsymbol{T}$ contains the nine unknown nodal temperatures, and $\boldsymbol{b}$ represents the known boundary-condition terms. Solving this system provides the temperature distribution throughout the composite wall. The resulting temperatures were independently verified using thermal resistance calculations, Fourier's law of heat conduction, and convection energy balances.

Original Problem Results

The original composite wall was analyzed using the finite difference method with nine equally spaced nodes. The resulting system of linear equations was solved to determine the steady-state temperature distribution across the stainless steel and nonmetallic layers. The results show a progressive decrease in temperature from the bottom stainless-steel surface, exposed to hot steam, to the upper nonmetallic surface, exposed to cooler air.

Nodal Temperature Distribution

Table 3. Nodal Temperatures for the Original Configuration

Node	Position, x (mm)	Temperature (°C)
0	0	232.39
1	5	229.20
2	10	226.02
3	15	222.83
4	20	219.65
5	25	216.46
6	30	213.27
7	35	199.47
8	40	185.66

The highest temperature occurs at node 0, while the lowest temperature occurs at node 8. The temperature decreases linearly within each material, with a change in slope at the interface due to the difference in thermal conductivity.

Steady-State Heat Flux

The total thermal resistance per unit area was calculated by adding the conduction and convection resistances in series:

$$R''_{total} = \frac{1}{h_1} + \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{1}{h_2}$$

Substituting the original problem parameters:

$$R''_{total} = \frac{1}{300} + \frac{0.030}{13} + \frac{0.010}{3} + \frac{1}{50}$$

$$R''_{total} = 0.02897436\ m^2 * K/W$$

The steady-state heat flux was determined using:

$$q'' = \frac{T_{\infty,1} - T_{\infty,2}}{R''_{total}}$$

$$q'' = \frac{260 - 20}{0.02897436}$$

$$q'' = 8{,}283.18\ W/m^2$$

These results were independently verified through manual calculations and were consistent with the AI-generated values within rounding differences.

Temperature Distribution

The temperature distribution across the original composite wall shows a continuous decrease from the steam-side surface to the air-side surface. The stainless steel layer exhibits a smaller temperature gradient due to its higher thermal conductivity, while the nonmetallic layer exhibits a steeper temperature gradient because of its lower thermal conductivity. At the interface (node 6), the temperature remains continuous under the assumption of perfect thermal contact.

Temperature Limit Evaluation

The maximum temperature of the stainless steel was 232.39 °C, which is below the specified limit of 260 °C. However, the temperature at the material interface was approximately 213.27 °C, exceeding the 204 °C limit specified for the ASTM B211 6061 bolts under the simplified temperature-comparison criterion. Because the one-dimensional model does not explicitly account for heat conduction through the bolts, this comparison does not establish their actual temperatures or complete compliance with the ASME code. These results serve as the baseline for evaluating the improvements introduced in the adapted configuration.

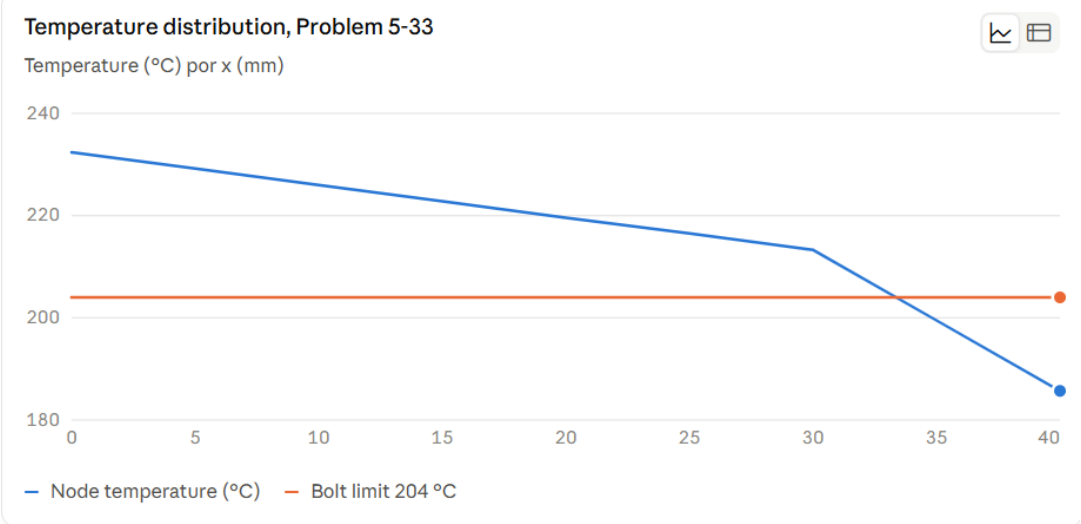


Figure 2. *Steady-State temperature distribution for the original composite wall.*

Adapted Problem and Engineering Justification

The original composite wall was modified to improve its thermal performance by increasing the steady-state heat transfer rate and reducing the operating temperatures of the materials. The adaptation was developed by changing two physical parameters: the thermal conductivity of the nonmetallic layer and the convection heat transfer coefficient at the upper surface. The geometry, stainless steel properties, steam temperature, and ambient air temperature were maintained to allow a direct comparison between the original and adapted configurations.

Proposed Modifications

Two modifications were introduced to the original problem:

1. The thermal conductivity of the nonmetallic material was increased from 3 to 12 W/m·K.

2. The convection heat transfer coefficient at the upper surface was increased from 50 to 150 W/m²·K.

Table 4. Original and Adapted Thermal Parameters

Parameter	Original	Adapted
Stainless steel thermal conductivity, k_1 (W/m·K)	13	13
Nonmetallic thermal conductivity, k_2 (W/m·K)	3	12
Stainless steel thickness, L_1 (mm)	30	30
Nonmetallic thickness, L_2 (mm)	10	10
Bottom convection coefficient, h_1 (W/m²·K)	300	300
Top convection coefficient, h_2 (W/m²·K)	50	150
Steam temperature (°C)	260	260
Air temperature (°C)	20	20

Nodal spacing (mm)	5	5

Thermal Conductivity Modification

The first modification involved increasing the thermal conductivity of the nonmetallic layer from 3 to 12 W/m·K. According to Fourier's law, a material with higher thermal conductivity offers less resistance to heat transfer when its thickness and surface area remain constant. The thermal resistance of a plane wall is expressed as:

$$R''_{cond} = \frac{L}{k}$$

Therefore, increasing thermal conductivity by a factor of four reduces the conduction resistance of the nonmetallic layer to one-fourth of its original value. This modification facilitates heat conduction through the second layer and contributes to an increase in the overall heat flux.

Convection Heat Transfer Modification

The second modification involved increasing the upper convection heat transfer coefficient from 50 to 150 W/m²·K. The thermal resistance associated with convection is:

$$R''_{conv} = \frac{1}{h}$$

Increasing the convection coefficient by a factor of three reduces the upper convection resistance to one-third of its original value. In practice, this modification could be approached by introducing forced airflow, such as fans or ducted ventilation, to increase convective heat transfer at the exposed surface.

Engineering Justification

- The proposed modifications were selected because the original configuration exhibited significant thermal resistance at the upper air-side boundary.
- Reducing both the conduction resistance of the nonmetallic layer and the convection resistance at the upper surface increases the steady-state heat flux through the composite wall.
- The adapted configuration also produces lower wall temperatures, potentially improving the thermal operating conditions of the materials.
- However, the proposed nonmetallic conductivity of 12 W/m·K represents an assumed material property. A suitable material would need to be identified and verified using manufacturer data, including its operating temperature limits and compatibility with the assembly.
- Similarly, achieving a convection coefficient of 150 W/m²·K would require further analysis of airflow conditions and surface geometry.

- The adaptation is therefore presented as a theoretical thermal improvement rather than a fully validated physical design.

Adapted Problem Results

The adapted composite wall was analyzed using the same finite difference method applied to the original configuration. The thermal conductivity of the nonmetallic material was increased to 12 W/m·K, and the upper convection heat transfer coefficient was increased to 150 W/m²·K. All remaining parameters, including the wall thicknesses, stainless steel thermal conductivity, fluid temperatures, and nodal spacing, were maintained constant. The updated finite difference equations were solved to determine the steady-state temperature distribution across the nine nodes.

Nodal Temperature Distribution

Table 5. Nodal Temperatures for the Adapted Configuration

Node	Position, x (mm)	Temperature (°C)
0	0	199.12
1	5	192.10
2	10	185.07
3	15	178.05
4	20	171.02
5	25	164.00
6	30	156.98
7	35	149.37
8	40	141.76

The maximum wall temperature occurs at node 0, while the minimum occurs at node 8. The temperature decreases continuously across the composite wall, with a slight change in slope at the material interface. The adapted configuration produces lower temperatures throughout both materials compared with the original design.

Adapted Steady-State Heat Flux

The total thermal resistance per unit area was calculated using the modified thermal properties:

$$R''_{total} = \frac{1}{h_1} + \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{1}{h_2}$$

Substituting the adapted values:

$$R''_{total} = \frac{1}{300} + \frac{0.030}{13} + \frac{0.010}{12} + \frac{1}{150}$$

$$R''_{total} = 0.01314103\ m^2 * K/W$$

The steady-state heat flux was determined using:

$$q'' = \frac{T_{\infty,1} - T_{\infty,2}}{R''_{total}}$$

$$q'' = \frac{260 - 20}{0.01314103}$$

$$q'' = 18{,}263.41\ W/m^2$$

The adapted heat flux is greater than the original value of 8,283.18 W/m², demonstrating that the proposed modifications reduce the overall thermal resistance and increase heat transfer through the composite wall.

Temperature Distribution

The adapted temperature profile shows a continuous decrease from the steam-side surface to the air-side surface. The stainless steel and nonmetallic layers exhibit nearly similar temperature gradients because their thermal conductivities are relatively close, at 13 and 12 W/m·K, respectively. A slight change in slope occurs at node 6, corresponding to the interface between the two materials. The results demonstrate that increasing thermal conductivity and upper-surface convection reduces the wall temperatures while increasing the steady-state heat flux.

Temperature Limit Evaluation

The maximum stainless-steel temperature in the adapted configuration was 199.12 °C, which is below the specified limit of 260 °C. The interface temperature was approximately 156.98 °C, which is also below the 204 °C limit specified for the ASTM B211 6061 bolts under the simplified temperature-comparison criterion. These results indicate an improvement in the thermal conditions of the composite wall. However, because the model does not explicitly account for heat conduction through the bolts, their actual temperatures and complete ASME code compliance cannot be established through this analysis alone
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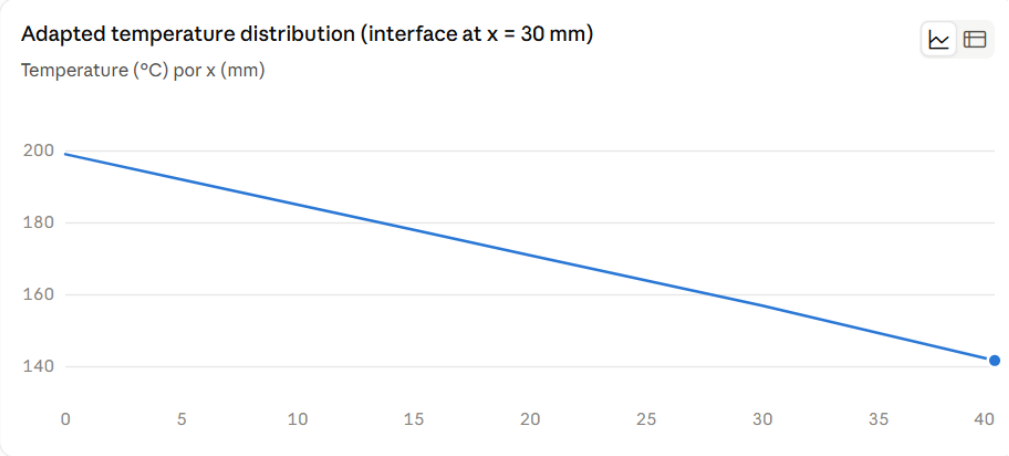


Figure 3. *Steady-State temperature distribution adapted*

Comparative Analysis

The original and adapted configurations were compared to evaluate the effects of increasing the thermal conductivity of the nonmetallic layer and the upper convection heat transfer coefficient. Both configurations were analyzed using the same finite difference method, geometry, fluid temperatures, and nodal spacing to ensure a consistent comparison.

Thermal Resistance and Heat Flux Comparison

Table 6. Comparison of Thermal Performance

Parameter	Original Configuration	Adapted Configuration
Nonmetallic thermal conductivity, k_2 (W/m·K)	3	12
Upper convection coefficient, h_2 (W/m²·K)	50	150
Total thermal resistance (m²·K/W)	0.02897435	0.01314103
Steady-state heat flux (W/m²)	8,283.18	18,263.41
Maximum wall temperature (°C)	232.39	199.12
Interface temperature (°C)	213.27	156.98
Minimum wall temperature (°C)	185.66	141.76

The adapted configuration reduced the overall thermal resistance from 0.02897435 to 0.01314103 m²·K/W. Consequently, the heat flux increased from 8,283.18 to 18,263.41 W/m².

The percentage increase in heat flux was determined using:

$$\%\ Increase = \frac{q''_{adapted} - q''_{original}}{q''_{original}} \times 100$$

$$\%\ Increase = \frac{18{,}263.41 - 8{,}263.18}{8{,}263.18} \times 100$$

$$\% \, Increase = 120.5$$

$$\text{Heat Flux Increase} \approx \ 120.5\%$$

This represents approximately 2.20 times the original steady-state heat flux.

Temperature Distribution Comparison

Table 7. Original and Adapted Nodal Temperature Comparison

Node	Position (mm)	Original (°C)	Adapted (°C)
0	0	232.39	199.12
1	5	229.20	192.10
2	10	226.02	185.07
3	15	222.83	178.05
4	20	219.65	171.02
5	25	216.46	164.00
6	30	213.27	156.98
7	35	199.47	149.37
8	40	185.66	141.76

The adapted configuration exhibits lower temperatures at every node. The maximum wall temperature decreased by approximately 33.27 °C, while the interface temperature decreased by approximately 56.30 °C. The temperature profiles remain continuous across the interface, while their slopes change according to the thermal conductivity of each material.

Engineering Interpretation

- The increase in heat flux is primarily associated with the reduction in conduction resistance through the nonmetallic material and convection resistance at the upper surface.

- Although the heat transfer rate per unit area increases, the wall temperatures decrease because the modified system transfers heat more effectively toward the cooler surrounding air.

- The adapted configuration also provides greater temperature margins relative to the specified material limits. However, further engineering analysis would be required to validate the proposed material and convection conditions under actual operating conditions.

- The results demonstrate an improvement in steady-state thermal performance, but they do not establish a reduction in transient cooling time.

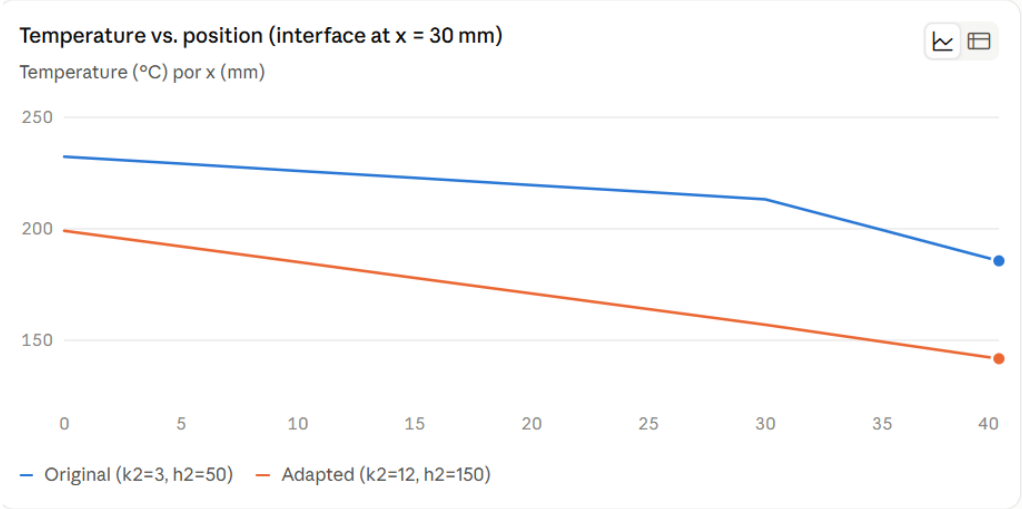


Figure 4. *Steady-State temperature distribution: adapted vs original*

Supplementary Transient Heat Transfer Analysis

The original and adapted configurations of Problem 5-33 were analyzed under steady-state conditions. The results demonstrated an approximately 120.5% increase in heat flux after increasing the thermal conductivity of the nonmetallic layer and the convection coefficient at the air-exposed surface. Although this improvement indicates a greater steady-state heat transfer rate per unit area, it does not directly establish the time required for the wall to heat up or cool down. For this reason, a supplementary transient heat transfer formulation was developed to identify the governing equations, determine how time-dependent temperatures would be calculated, and explain the additional information required for a quantitative comparison.

Governing Transient Heat Conduction Equation

For one-dimensional transient conduction without internal heat generation, the energy equation is:

$$p \cdot c_p \frac{dT}{dt} = \frac{d}{dx}\left(k \frac{dT}{dx}\right)$$

Assuming constant thermal conductivity within each homogeneous layer:

$$p \cdot c_p \frac{dT}{dt} = k \frac{d^2T}{dx^2}$$

Dividing both sides by the volumetric heat capacity:

$$\frac{dT}{dt} = \frac{k}{p \cdot c_p} \cdot \frac{d^2T}{dx^2}$$

The thermal diffusivity is defined as: $\alpha = \frac{k}{p \cdot c_p}$

Therefore:

$$\frac{dT}{dt} = \alpha \cdot \frac{d^2T}{dx^2}$$

where:

* T = temperature (°C)
* t = time (s)
* k = thermal conductivity (W/m·K)
* ρ = density (kg/m³)
* cp = specific heat capacity (J/kg·K)
* α = thermal diffusivity (m²/s)

Thermal diffusivity describes how quickly temperature disturbances propagate through a material relative to its capacity to store thermal energy.

Transient Finite Difference Formulation

The same nine-node discretization used in the steady-state analysis is maintained, with a nodal spacing of:

$$\Delta x = 0.005\, m$$

The time derivative is approximated using a forward difference:

$$\frac{dT_i}{dt} \approx \frac{T_i^{n+1} - T_i^n}{\Delta t}$$

The second spatial derivative is approximated by:

$$\frac{d^2T_i}{dx^2} \approx \frac{T_{i-1}^n - 2T_i^n + T_{i+1}^n}{(\Delta x)^2}$$

Substituting into the transient conduction equation:

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha_j \frac{T_{i-1}^n - 2T_i^n + T_{i+1}^n}{(\Delta x)^2}$$

Rearranging to calculate the temperature at the next time step:

$$T_i^{n+1} - T_i^n = \frac{\alpha_j \Delta t}{(\Delta x)^2}(T_{i-1}^n - 2T_i^n + T_{i+1}^n)$$

Defining the Fourier number:

$$Fo_j = \frac{\alpha_j \Delta t}{(\Delta x)^2}$$

The explicit finite difference equation becomes:

$$T_i^{n+1} = T_i^n + Fo_j(T_{i-1}^n - 2T_i^n + T_{i+1}^n)$$

This expression applies to interior nodes located entirely within one homogeneous material. The convection boundaries and material interface require separate energy balances.

Transient Energy Balances for the Nine Nodes

For convenience, define the volumetric heat capacities of the two materials:

$$C_1 = p_1 c_{p,1}$$

$$C_2 = p_2 c_{p,2}$$

Node 0 – Steam-side boundary

$$C_1 \cdot \frac{\Delta x}{2}\frac{dT_0}{dt} = h_1\left(T_{\infty,1} - T_0\right) + \frac{k_1}{\Delta x}(T_1 - T_0)$$

Nodes 1–5 – Stainless steel interior

For i = 1, 2, 3, 4, and 5:

$$C_1 \cdot \Delta x \frac{dT_i}{dt} = \frac{k_1}{\Delta x}\,(T_{i-1} - 2T_i + T_{i+1})$$

Node 6 – Material interface

$$\left(C_1 \cdot \frac{\Delta x}{2} + C_2 \cdot \frac{\Delta x}{2}\right)\frac{dT_6}{dt} = \frac{k_1}{\Delta x}\,(T_5 - T_6) + \frac{k_2}{\Delta x}(T_7 - T_6)$$

Node 7 – Nonmetallic interior

$$C_2 \cdot \Delta x \frac{dT_7}{dt} = \frac{k_2}{\Delta x}(T_6 - 2T_7 + T_8)$$

Node 8 – Air-side boundary

$$C_2 \cdot \frac{\Delta x}{2} \cdot \frac{dT_8}{dt} = \frac{k_2}{\Delta x}(T_7 - T_8) + h_2\left(T_{\infty,2} - T_8\right)$$

These nine energy balances account for conduction between adjacent nodes, convection at the exposed surfaces, and thermal energy storage within the composite wall. At steady state, all time derivatives become zero, reducing the governing balances to the steady-state equations used in the original numerical analysis.

Matrix Formulation

The nine transient equations can be expressed in matrix form:

$$M\,\frac{dT}{dt} + KT = f$$

where M is the thermal capacity matrix, K is the thermal conductance matrix, T is the nodal temperature vector, and f contains the fluid-temperature boundary contributions.

For constant boundary conditions, the steady-state solution satisfies:

Defining the temperature deviation from steady state:

$$\theta(t) = T(t) - T_{ss}$$

Multiplying by the inverse thermal capacity matrix:

$$\frac{d\theta}{dt} = -M - 1 \cdot K\theta$$

For constant coefficients, the solution is:

$$\theta(t) = e^{-M^{-1 \cdot Kt}}\,\theta$$

This equation describes the transient temperature response of the complete nine-node system.

Mathematical Derivation of Time

To illustrate how time may be determined, consider an individual exponential thermal mode represented by:

$$\frac{d\theta}{dt} = -\lambda\,\theta$$

Separating variables:

$$\frac{d\theta}{\theta} = -\lambda\,dt$$

Integrating between the initial condition and a later time:

$$\int_{\theta_0}^{\theta(t)} \frac{d\theta}{\theta} = -\lambda \int_0^t dt$$

Solving for time:

$$\ln\left|\frac{\theta(t)}{\theta_0}\right| = -\lambda t$$

Since:

$$\theta = T - T_{ss}$$

The time expression for a single thermal mode becomes:

$$t = \frac{1}{\lambda}\ln\left|\frac{T_{inicial} - T_{ss}}{T_x - T_{ss}}\right|$$

This expression is exact for a single exponential thermal mode. It must not be treated as the general exact solution for an arbitrary node of the nine-node composite wall. The complete system has multiple thermal modes. Its characteristic decay rates are obtained from:

$$\det(K - \lambda M) = 0$$

For each node, the general solution can be written as:

$$T_i(t) - T_{i.ss} = \sum_{m=1}^{9} A_{i,m} e^{-\lambda_m t}$$

For a specified target (x) temperature, time is determined from:

$$\sum_{m=1}^{9} A_{i,m} e^{-\lambda_m t} = T_x - T_{i,ss}$$

In general, this equation must be solved numerically after the material properties, initial temperatures, and response criterion are established.

Missing Material Properties and Data Limitations

Although the original problem provides wall thicknesses, thermal conductivities, convection coefficients, and surrounding fluid temperatures, the available information is insufficient to calculate a unique transient heating or cooling time.

The following information would be required:

1. Density of the stainless steel and nonmetallic materials.
2. Specific heat capacity of both materials.
3. Initial temperature distribution throughout the wall.
4. A specific target temperature or transient response criterion.
5. The thermal boundary conditions during the heating or cooling process.

The adapted nonmetallic layer was defined through an assumed thermal conductivity of 12 W/(m·K). A specific material and its corresponding density and specific heat capacity would also need to be identified and validated before performing a numerical transient simulation. Without these data, thermal diffusivities, thermal capacity matrix values, decay constants, and numerical response times cannot be determined reliably.

Engineering Interpretation

The adapted steady-state configuration increases the calculated heat flux from approximately 8,283.18 W/m² to 18,263.41 W/m², representing an improvement of approximately 120.5%. This result demonstrates a reduction in overall thermal resistance and an increase in steady-state heat transfer capability under the specified boundary conditions. However, an increase in steady-state heat flux does not necessarily correspond to the same percentage reduction in heating or cooling time, because transient response also depends on the density and specific heat capacity of each material. The supplementary transient formulation establishes the mathematical basis for a future heating or cooling time comparison while explicitly recognizing the limitations of the information provided in the selected textbook problem. Therefore, the improvement established by the present project is an increase in steady-state heat transfer performance, rather than a numerically verified reduction in cooling time.

AI Output Audit and Manual Verification

Artificial intelligence was used as a supporting tool to solve the original and adapted versions of Problem 5-33. Structured prompts were developed to guide Claude through the finite difference formulation, boundary conditions, material interface analysis, matrix assembly, and numerical calculations. To evaluate the reliability of the AI-generated solutions, independent manual audits were performed for both configurations. The calculations were reviewed using fundamental heat transfer principles, including Fourier's law, convection heat transfer equations, thermal resistance analysis, and conservation of energy.

Original Problem Manual Audit

- The original problem was independently verified by calculating the total thermal resistance, steady-state heat flux, and temperatures at all nine nodes.
- The manual calculations produced a total thermal resistance of 0.02897435 m²·K/W and a heat flux of approximately 8,283.18 W/m².
- The nodal temperatures were verified using Fourier's law and finite difference equations. Energy conservation was also checked at the interface between the stainless steel and nonmetallic materials, as well as at the convection boundaries.
- The calculated temperatures were consistent with Claude's results within rounding differences.
- The audit also identified a limitation in the simplified temperature evaluation: the interface temperature exceeded the specified bolt temperature limit, but the actual bolt temperatures could not be determined using the one-dimensional model.

Adapted Problem Manual Audit

- The adapted configuration was independently verified following the same engineering principles and numerical methodology.

- The manual calculations confirmed a total thermal resistance of 0.01314103 m²·K/W and a steady-state heat flux of approximately 18,263.41 W/m².
- The temperatures at all nine nodes were verified, and the interface energy balance and convection boundary conditions were evaluated.
- The resulting temperatures agreed with Claude's calculations within rounding differences.
- The audit also confirmed that the adapted configuration increased the steady-state heat flux by approximately 120.5 % compared with the original configuration.

AI-Generated Errors and Corrections

During the verification process, numerical and interpretative inconsistencies were identified in the AI-generated output.

Table 8. AI Output Verification and Corrections

Observation	AI Output	Manual Verification
Rounding in convection calculation	300*(260-199.12) = 18,264 W/m^2	Using rounded values gives 18,264 W/m^2; using unrounded temperature gives approximately 18,263.41 W/m^2
Adapted temperature gradient	Nonmetallic layer described as having a smaller temperature gradient	The nonmetallic layer has a slightly larger temperature gradient in magnitude than the stainless steel
Bolt Temperature assessment	Modeled wall temperatures compared with bolt limits	Actual bolt temperatures cannot be established without additional analysis
Numerical accuracy	Nodal temperatures reported with limited decimal places	Manual calculations confirmed agreement within rounding differences.

Documentation and Verification Process

- The prompts and AI-generated outputs for both configurations were documented using GitHub to maintain a record of the problem-solving process.
- The repository includes the original and adapted prompts, AI responses, and supporting manual calculation audits.
- The manual verification documents provide evidence of the mathematical procedures used to confirm the numerical solutions and identify limitations or inconsistencies.
- Both students reviewed the calculations as part of the engineering verification process. The signed certification accompanying the final report will document their review of the submitted work.
- The complete handwritten verification procedures for both the original and adapted configurations are included in the supporting audit documents available in the project GitHub repository.

*Manual-Audit-Original-5-33.pdf

* Manual-Audit-Adapted-5-33.pdf

Conclusions

This project evaluated the thermal performance of a composite wall through the finite difference method, combining artificial intelligence assistance with independent engineering verification. The original and adapted configurations of Problem 5-33 were analyzed under steady-state, one-dimensional heat conduction conditions to determine the temperature distribution across nine nodes and evaluate the effects of changes in material properties and convection conditions.

For the original configuration, the total thermal resistance was 0.02897435 $m^2 \cdot K/W$, corresponding to a steady-state heat flux of approximately 8,283.18 W/m^2. The maximum wall temperature was 232.39 °C, while the temperature at the interface between the two materials was 213.27 °C. The interface temperature exceeded the specified bolt temperature limit under the simplified comparison criterion, highlighting the importance of evaluating thermal conditions within the assembly.

The adapted configuration was developed by increasing the thermal conductivity of the nonmetallic layer from 3 to 12 $W/m \cdot K$ and the upper convection heat transfer coefficient from 50 to 150 $W/m^2 \cdot K$. These modifications reduced the overall thermal resistance to 0.01314103 $m^2 \cdot K/W$ and increased the steady-state heat flux to approximately 18,263.41 W/m^2, representing an improvement of 120.5%. Furthermore, the maximum wall temperature decreased to 199.12 °C, while the interface temperature decreased to 156.98 °C, demonstrating improved thermal performance under the modeled conditions.

To complement the steady-state analysis, a supplementary transient heat transfer formulation was developed. This analysis extended the nine-node finite difference model by incorporating thermal energy storage, transient conduction equations, and a matrix representation of the time-dependent temperature response. A mathematical expression for time was derived for an individual exponential thermal mode, while the complete wall was represented as a system involving multiple thermal modes.

The transient analysis also identified limitations in the available problem data. Material densities, specific heat capacities, initial temperature distributions, and a defined response criterion are necessary to determine numerical heating or cooling times. Consequently, although the adapted configuration demonstrates a substantial improvement in steady-state heat transfer capability, a numerical reduction in cooling time cannot be established from the available information alone.

Independent manual audits verified the numerical results generated with Claude and identified minor rounding and interpretation inconsistencies. These findings reinforced the importance of critically evaluating artificial intelligence outputs rather than relying exclusively on automated calculations. The prompts, generated responses, manual verifications, and supporting documentation were organized in GitHub to maintain a transparent record of the engineering process.

Overall, the project demonstrates how numerical methods, artificial intelligence, and independent verification can be integrated to evaluate and improve thermal system performance. The proposed adaptation provides a significant increase in steady-state heat flux, while the supplementary transient formulation establishes the mathematical foundation for future time-dependent analysis once the necessary material properties and initial conditions become available.

References

[1] Çengel, Y. A., & Ghajar, A. J. (2020). Heat and Mass Transfer: Fundamentals & Applications (6th ed.). McGraw-Hill Education. Chapter 5, Problem 5-33.

[2] Anthropic. (2026). Claude [Large language model]. https://claude.ai/

[3] GitHub. (2026). Heat-Transfer-Project-5-33 [Project repository]. https://github.com/valeriasantiago35-arch/Heat-Transfer-Project-5-33

Engineering Verification and Certification

We, the undersigned students, certify that we have independently reviewed and manually verified the AI-generated numerical calculations, finite difference equations, boundary conditions, thermal resistance calculations, and temperature distributions presented in this report for both the original and adapted versions of Problem 5-33. We further certify that the AI-generated solutions were evaluated using fundamental heat transfer principles, including Fourier's law, convection heat transfer, and conservation of energy. Any identified numerical inconsistencies, corrections, and modeling limitations have been documented. The supporting manual audits, AI prompts, and generated outputs are included in the project's GitHub repository.

By signing below, we confirm that we have personally reviewed the calculations and verification procedures documented in this report.

Student 1

Full Name: Valeria L Santiago Maldonado

Signature: 

Date: 24/10/2026

Student 2

Full Name: Christian Figueroa Coriano



Signature:

Date: 24/10/2026